

A series of four parallel white diagonal lines of varying lengths, starting from the left edge and extending towards the center of the slide.

DAY 3 AGENDA

- ▶ Roles
- ▶ Handlers
- ▶ Templates “.j2”
- ▶ Ansible-Vault
- ▶ Ansible-Galaxy

ROLES

Roles: pack related vars, files, templates, tasks, and handlers, based on a known file structure so you can easily reuse them and share them with other users.



main.yml



main.yml



- ▶ Create your first role with name (web)
- ▶ The task book will include:
 1. installing a package
(get the package name from vars)
 2. copying a list of files from controller to host using loop
(get the list of file names from vars)
(the actual files will be stored in ./roles/web/files)
(will be executed only when the install task is in state: changed)
- ▶ Restart the service of the installed package
(will be executed only when the copy task is in state: changed)

ROLES



HANDLERS

Handlers: Handlers are tasks that only run when notified.

Example:

- name: my play with handlers

hosts: all

tasks:

- name: my task1 with handlers

apt:

name: nginx

notify: my_handler

handlers:

- name: my_handler

service:

name: nginx

state: restarted



main.yml



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- ▶ Create your first role with name (web)
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 1. installing a package
(get the package name from vars)
 2. copying a list of files from controller to host using loop
(get the list of file names from vars)
(the actual files will be stored in ./roles/web/files)
(will be executed using Handlers)
- ▶ Restart the service of the installed package
(will be executed using Handlers chaining)

HANDLERS



TEMPLATES

Templates: Ansible uses Jinja2 templates to create dynamic content at the controller end, and render it as static content at the host end.

index.html.j2

```
<html>
<body>
<h1> {{ my_message }} <h1>
</body>
</html>
```

index.html

```
<html>
<body>
<h1> Hello from ansible <h1>
</body>
</html>
```



main.yml



main.yml



main.yml



Index.html.j2

- ▶ Create your first role with name (web)
- ▶ The task book will include:
 1. installing a package
(get the package name from vars)
 2. Copying a file from controller to host using template
(get the template name & template message from vars)
(the actual template file will be stored in `./roles/web/templates`)
(will also notify the restart handler)
 3. copying a list of files from controller to host using loop
(get the list of file names from vars)
(the actual files will be stored in `./roles/web/files`)
(will be executed using Handlers)
- ▶ Restart the service of the installed package
(will be executed using Handlers chaining)

TEMPLATES



ANSIBLE-VAULT

Ansible-Vault: provides a way to encrypt and manage sensitive data such as passwords.

Example:

```
- name: my play with vault
hosts: all
vars:
  user_name: ahmed
var_files:
  - ./passwords.yml
tasks:
- name: my task1 with vault
  user:
    name: "{{ user_name }}"
    password: "{{ user_pass }}"
```

passwords.yml

user_pass: 123456

Commands:

```
$ ansible-vault encrypt ./passwords.yml
```

```
$ ansible-playbook my-playbook.yml --ask-vault-pass
```

ANSIBLE-GALAXY

Ansible-Galaxy: refers to the Galaxy website, a free site for finding, downloading, and sharing community developed roles and collections.

Commands:

```
$ ansible-galaxy info username.role_name
$ ansible-galaxy install username.role_name
$ ansible-galaxy install username.role_name,1.0.0
$ ansible-galaxy list
$ ansible-galaxy remove username.role_name
```